Assignment 1BM130

1BM130 - quartile 4 (2025-2026)

Group number 5

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1 | Management summary

Sustainable urban mobility is a central challenge for Dutch municipalities. As cities are required to develop Sustainable Urban Mobility Plans, there is a growing demand for tools that connect cycling accessibility data to actual travel behaviour and help municipalities decide where different policy interventions are needed. This report addresses that demand through three integrated analyses, carried out in collaboration with SmartwayZ and based on the ODiN 2024 national travel survey, the CBS 2024 neighbourhood-level socio-economic dataset, and the SmartwayZ buurt-to-buurt routing dataset.

The first analysis examines where the 10-minute cycling city performs well or poorly across Dutch municipalities, and how cycling access relates to actual local cycling behaviour. A weighted amenity access score (Bike-10) was constructed for each neighbourhood, covering fourteen essential amenity categories, including supermarkets, GP services, primary schools, and pharmacies. The results show that highly urbanized municipalities, such as Haarlem, Leiden, Delft, and Amsterdam, score highest on cycling access, while rural municipalities score lowest. Importantly, high access does not automatically translate into high local cycling usage. A clear access-usage mismatch was identified across many municipalities, with cities such as Heerlen, Schiedam, and Eindhoven showing high access but below-average cycling rates. The province of Gelderland showed the highest share of local cycling trips within the 10-minute bike shed, while Groningen performed lowest. The analysis also found that, contrary to common assumptions, low-income residents tend to have equal or slightly higher cycling access than higher-income groups, likely due to their concentration in denser urban neighbourhoods. However, their overall cycling behaviour is more strongly shaped by urbanisation context than by income alone.

The second analysis developed a predictive model for transport mode choice at the trip level, distinguishing between car, public transport, bike, and walking trips. XGBoost performed best, with a final test accuracy of 80.5% and a macro F1-score of 0.78. The model performed well for car, bike, and walking trips, while public transport was more difficult to predict. The most important predictors were trip distance, trip purpose, and reported travel habits. Amenity availability contributed additional information but ranked lower than behavioural and trip-related variables. This supports the conclusion that improving cycling access is important but not sufficient on its own. Travel habits, trip characteristics, and perceived accessibility also matter.

The third analysis translated these findings into an interactive Streamlit dashboard for policymakers. The dashboard allows users to select any Dutch municipality, inspect its Bike-10 access score and low-income context, and explore illustrative what-if scenarios, such as adding a supermarket or improving cycling accessibility by 10-20%. A Gemini-powered AI assistant is embedded in the dashboard to explain the selected results and provide concise, equity-focused policy guidance based on the selected municipality and scenario.

Together, the three analyses show that the 10-minute cycling city is a useful framework for the Dutch context, but it should be used as a diagnostic tool rather than as a simple accessibility ranking. The main recommendation to SmartwayZ is to distinguish among policy cases: low-access municipalities may need improved access to essential amenities, while high-access but low-usage municipalities may need interventions that address travel habits, perceived accessibility, and the attractiveness of sustainable transport options. Because the analyses are based on cross-sectional data, the results should be interpreted as associations and policy opportunities rather than proven causal effects.

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2 | General Introduction

Sustainable urban mobility is one of the central challenges facing Dutch municipalities today. As cities are required to develop Sustainable Urban Mobility Plans, there is a growing need to understand not only where essential services are located, but how residents actually travel to them, and why. For municipalities, this is important because these plans require practical policy perspectives. The Netherlands, despite having the world's most extensive cycling network, still sees a significant share of short-distance trips made by car.

This report investigates the concept of the 10-minute cycling city for the Dutch context: the idea that key daily amenities should be reachable within a 10-minute cycling radius. This project is carried out in collaboration with SmartwayZ, with the goal of developing insights and a practical dashboard that help municipalities see where cycling access is strong, where cycling usage remains behind, and which factors are related to sustainable mode choice. The analysis is based on three datasets: the ODiN 2024 national travel survey, the CBS 2024 neighbourhood-level socio-economic dataset, and the SmartwayZ buurt-to-buurt routing dataset.

The report is structured in three parts. Section 1 presents a descriptive analytics analysis of cycling access and usage across Dutch municipalities. Section 2 develops a predictive model for transport mode choice, providing the analytical foundation for the SmartwayZ tool. Section 3 describes the implementation of this tool as an interactive dashboard, incorporating AI-driven insights to support municipal policymakers in exploring and acting on the results. Section 4 integrates the findings, discusses the strengths and limitations of the approach, and presents recommendations for SmartwayZ and municipalities.

3 | Descriptive Analytics

3.1 | Introduction

The Netherlands is a cycling country, with the largest bike network in the world. Yet many people still don't cycle even for short trips. Cities already know how accessible the amenities are, but they lack insight into how this relates to the usage. They have to create a Sustainable Urban Mobility Plan, where they must address this relationship. It is unknown whether good cycling access actually leads to cycling behaviour, and how this varies across neighbourhoods and socio-economic groups. For example, it could be that people with low incomes are unable to afford houses in the city center and are therefore further away from amenities. A 10-minute cycling radius may reduce segregation and improve access, but this has never been quantified for the Netherlands. Therefore, in this search, insights are gained into how many essential amenities are actually reachable within a realistic cycling radius, whether people truly use the bicycle when access is good, which groups benefit or are left behind, and whether a 10-minute cycling radius can reduce socio-economic segregation. Furthermore, it is researched how to improve the access and usage within municipalities, by comparing the good and poorly performing municipalities. Thus, there is a need to measure and compare cycling access, cycling usage, and social exposure to understand where the mobility system is effective, and where it is not.

3.2 | Research questions

To address the identified problem, two research questions have been set up and are discussed. The relevance of them is also stressed. This project is carried out in collaboration with SmartwayZ, with the goal of generating actionable insights for municipalities developing a Sustainable Urban Mobility Plan (SUMP).

1 - Where in the Netherlands does the 10-minute cycling city perform well or poorly in terms of access and local cycling usage, and which municipalities are most relevant for policy intervention?

This research question forms the descriptive core of the project. Its purpose is to establish whether the 10-minute cycling city is a meaningful concept in the Dutch context: not only as a spatial framework, but as a lens for understanding actual travel behaviour and socioeconomic equity. For the assignment, this question directly addresses the SmartwayZ objective of generating actionable insights for policymakers by identifying where the local cycling city works, where it falls short, and where intervention would have the greatest impact. Learning from the cities that perform well can give insights into possible ideas on how to improve them for cities that perform poorly.

2 - How do different income groups benefit equally from 10-minute cycling access to essential amenities, and how does urbanisation affect their local cycling behaviour?

In the Dutch context, cycling is often assumed to be universally accessible, yet this assumption masks structural differences between income groups, housing markets, and patterns of urban development. RQ2 addresses the equity dimension of the 10-minute cycling city by examining for whom good cycling access translates into real opportunities. Whereas RQ1 identifies where the system performs well or poorly, RQ2 investigates whether the mobility system reinforces or reduces socio-economic disparities.
Lower-income residents are more likely to live in areas with fewer amenities close-by, meaning a 10-minute cycling radius may offer fewer opportunities than in dense urban centers. At the same time, these groups often rely more heavily on affordable modes such as cycling. Understanding whether good access leads to actual cycling behaviour, and whether this differs across levels of urbanisation, is therefore essential.
For SmartwayZ, this relationship helps identify mismatches between access and usage, such as neighbourhoods with high access but low cycling rates, or areas where low-income groups consistently face poorer access despite high cycling potential. These insights support targeted interventions in amenity provision, cycling infrastructure, and integrated mobility–housing planning.
In the end, RQ2 ensures that the 10-minute cycling city is not only efficient but also socially just, revealing whether it can reduce socio-economic segregation or whether additional measures are needed to ensure equal benefits for all residents.

3.2.1 | Methods

To find the answer to the research questions, steps were taken, including the making of an isochrone for the 3km radius. combining data sources, measuring access to amenities within the isochrone, comparing access with local cycling behaviour, inspecting differences by income group, urbanisation class, municipality, and province. And finally, identifying policy-relevant access-usage patterns.

The following datasets were prepared for the data analysis: the ODiN 2024 dataset, the CBS 2024 dataset, the neighbourhood-to-neighbourhood routing from SmartwayZ (buurt-to-buurt), and the amenities locations.

The ODiN dataset consisted of the national weighted mobility survey of Dutch residents aged 6+. Where respondents report their travel for one assigned day. And therefore, it provides travel behaviour. Key attributes include trip distance, transport mode, trip purpose, and socio-demographic characteristics such as age, gender, and household income decile.

The CBS 2024 dataset is a neighbourhood-level dataset with spatial and socioeconomic indicators. Key attributes include urbanisation class, share of low-income residents, municipality and province labels, and population counts. The CBS 2024 dataset was chosen to use, because the socioeconomic indicators included a lot of missing values in the 2025 dataset, and it was assumed that these values would be quite similar to those of 2025, so the advice that would come out of this analysis would not be negatively influenced by this choice. The ODiN data was linked to the CBS data using the postal codes, the four digits. The resulting merged dataset contains approximately 17,400 trip records, where each row represents a single trip linked to its origin neighbourhoods and associated CBS socio-economic attributes. Together, these datasets connect cycling behaviour with spatial accessibility. The ODiN-based data describes whether people actually use bicycles and e-bikes for essential trips within a 10-minute radius, while the CBS-based data show where essential amenities are more or less accessible and where policy interventions may be most useful.

Furthermore, the SmartwayZ (neighbourhood-to-neighbourhood Routing) dataset provides travel distances between neighbourhoods and was used together with CBS data and the ODiN to create the amenity access score. For creating the access score, 7 essential amenity categories (e.g., Hospitals, Education, Supermarket) were identified, as specified by Kruijf (2026), using point location data from the amenities dataset provided by SmartwayZ.

To validate the isochrone definition of whether the 3 km radius and 10-minute threshold are empirically supported by actual cycling behaviour in the Netherlands, the distribution of cycling trip durations and distances (within_10min, dist_km) was examined to determine what share of trips falls within the proposed bike-shed. If the threshold captures a large share of cycling trips, the 10-minute cycling city is a meaningful descriptive framework; if most trips already exceed this distance, the concept requires recalibration for the Dutch context. The variables used for the recalibration are listed in Table B.1 in Section B.

The accessibility scores were weighted per category reachable within the isochrone per neighbourhood. The assigned weighting scores are shown in Table 3.1.

These values were assigned based on the relative importance of each amenity for daily essential needs, as defined by Kruijf (2026) amenities needed most frequently, such as supermarkets and GP services, received the highest weights, while less regularly visited services received lower weights. This allows for a meaningful differentiation in performance across neighbourhoods. This examines whether a higher weighted access score (bike10_weighted_score) is associated with a higher share of local cycling trips (pct_within_3km). This is the Dutch equivalent of the core finding in Abbiasov et al., 2024, where access explained 74–84% of local trip variation across US cities. If a comparable relationship holds in the Netherlands, it supports the use of access scores as a proxy for expected cycling behaviour and justifies access-based policy interventions.
Then, for each neighbourhood, the access score is computed. This shows whether each essential amenity category is reachable within the 10-minute cycling radius. Categories include: supermarkets, primary schools, GP/doctor, pharmacies, parks, cultural amenities, and services. As these were given by Kruijf (2026) to be referred to as the essential amenities. A binary indicator per category (1 = reachable, 0 = not reachable) is made based on whether there were 0, 1 or 2+ amenities within a certain neighbourhood. For each selected essential amenity category, a neighbourhood receives a value of 1 if the amenity is available

Table 3.1: Amenity Weights

Amenity	Weight
klasse_supermarkt	0.18
klasse_huisarts	0.14
klasse_basisschool	0.13
klasse_ziekenhuis	0.10
klasse_apotheek	0.07
klasse_voortgezet_onderwijs	0.07
klasse_bushalte	0.06
klasse_treinstation	0.05
klasse_horeca	0.05
klasse_restaurant	0.04
klasse_sportterrein	0.04
klasse_kinderopvang	0.03
klasse_fastfood	0.02
klasse_kledingwinkel	0.02

within the 10-minute cycling radius and 0 otherwise. The final score is calculated as the average of these binary indicators, meaning that a score of 1 represents access to all selected amenities, while a score of 0 represents access to none of them. This helps identify whether 10-minute cycling access is broadly available or whether some neighbourhoods have much weaker access to essential services.

Then the following variables were created to show the results of the access for the bike and walking modes, shown in Table 3.2. These variables describe the spatial side of the research question. They indicate whether essential amenities are available within the local cycling radius and allow comparison with the smaller walking-based 15-minute city definition. The travel behaviour variables describe how people actually move. They are used to inspect whether nearby essential amenities are associated with short-distance cycling and whether local trips are made by active modes or by car.

Table 3.2: Accessibility variables

Variable	Description
`bike10_weighted_score`	Weighted access index from 0 to 100 for essential amenities within the 10-minute cycling radius.
`bike10_score`	Count of how many essential amenity categories are reachable within the 10-minute cycling radius.
`bike10_klasse_*`	Individual amenity access indicators showing whether specific amenity types are reachable.
`walk15_weighted_score`	Comparable access index for the 15-minute walking radius, used as a reference to the original 15-minute city concept.

Relevant variables for research question 1
As it is important to gain insights into which municipalities are performing well and which are performing poorly, some geographical variables were defined, and shown below.

Table 3.3: Geographic variables used for RQ1

Variable	Description
`mismatch_gap`	Difference between a municipality's access rank and usage rank. A positive gap indicates high access but low usage (a policy opportunity where infrastructure or behaviour change is needed). A negative gap indicates cycling behaviour exceeds what access scores would predict.
`bike10_direct_gain_pct`	Projected percentage gain in access score if cycling routes were straightened to crow-fly distance. Identifies municipalities where detour factors limit amenity access, and where infrastructure improvements (more direct routes) would have the greatest effect.
`gm_naam / prov_label`	Municipality and province names, used to aggregate neighbourhood-level results to the spatial scale relevant for policymakers.
`ste_mvs (stedelijkheid)`	CBS urbanisation class (1 = very urban, 5 = rural). Used to contextualize access and usage results, since both variables are strongly shaped by urban density.

Relevant variables for research question 2
The income variables are used to inspect whether access and local cycling behaviour differ across socioeconomic groups. This is important because the 10-minute cycling city should not only provide nearby amenities, but should also do so equitably.

Table 3.4: Income and Context variables used for RQ2

Variable	Description
`p_ink_li`	Percentage of low-income residents in the neighbourhood.
`income_quintile`	Five neighbourhood income groups derived from `p_ink_li`.
`HHGestInkG`	Individual standardized household income decile in ODiN.
`income_group`	Grouped income bands used for comparison in the graphs.
`stedelijkheid_label`	urbanisation class, ranging from very urban to rural.
`gm_naam`	Municipality name.
`prov_label`	Province.
`age_group`	Age group of travellers.
`bike_type`	Bicycle, e-bike, or speed pedelec.

The second research question focuses on whether having essential amenities within a 10-minute cycling radius is associated with local travel behaviour, and whether this relationship differs across income groups and spatial contexts. Therefore, the analysis uses four groups of variables: accessibility variables, travel behaviour variables, income variables, and context variables.

The context variables are included because access and cycling behaviour are not only shaped by income or distance. urbanisation, municipality, age, and bike type can influence whether residents are able or willing to use cycling for short essential trips.

3.2.2 | Results

To identify whether the 3 kilometers radius for a bike would be useful, Figure 3.1 was created. This figure supports the use of distance as a central variable in the analysis: walking trips are concentrated at very short distances, cycling trips are still strongly represented within the local range, and car trips are distributed over longer distances. The following Figure 3.2 shows that there are differences in the willingness of bikers to travel to amenities with varying distances. This is split into two categories: the use of E-bikes and the use of regular bikes.

It is interesting to note that people are more willing to travel a longer distance when riding an e-bike than a regular bike. This could be because e-bikes are estimated to be approximately 25% faster than regular bikes (Kruijf, 2026). It is, however, not a constant difference; for example, going to education is

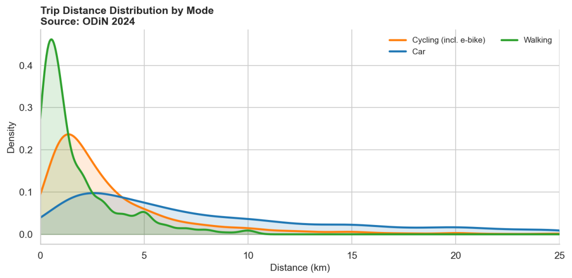


Figure 3.1: Trip Distance Distribution by Mode

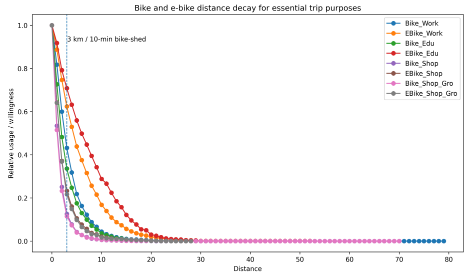


Figure 3.2: Share of essential cycling trips within 10-minute

highly impacted by whether someone can use an electric bike or not. But going shopping for groceries is impacted far less by this. Furthermore, it contributes by showing how cycling behaviour changes when the trip distance increases. The comparison between regular bicycles and e-bikes is important because e-bikes may extend the practical range of active mobility beyond the 10-minute threshold.

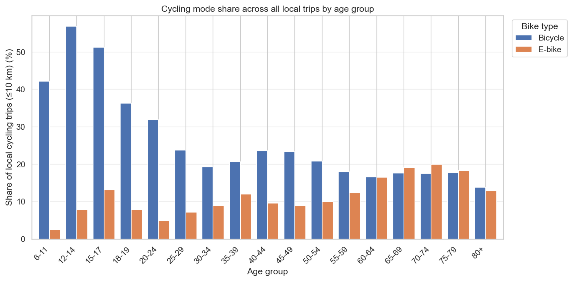


Figure 3.3: Bike type per age group

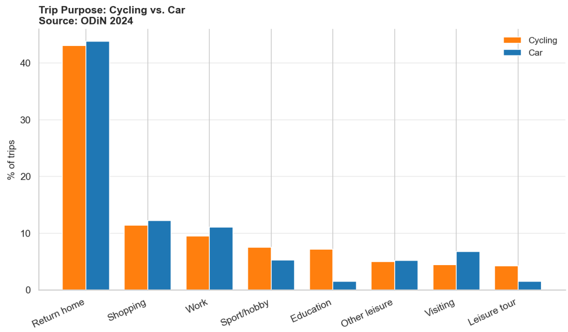


Figure 3.4: Trip purpose cycling vs car

The younger the person is, the more likely it is that he or she is using a regular bicycle currently, as shown in Figure 3.3. Finding this result highlights the importance of identifying whether, for example, education is within 3km. Mostly, the younger people have to go to school by bike. Furthermore, it can be seen that the bike is far more often taken to education than the car is, see Figure 3.4. As this is one of the identified essential amenities, this highlights the importance of accessible education by bike.

Results specifically related to research question 1

Research question 1 focuses on the cities that are performing well and poorly in terms of the access-usage relationship, to identify which municipalities are most relevant for policy intervention and how to do so.

Geographic variation across Dutch cities is examined. The relationship between access and usage may vary across provinces and municipalities. This is important for the SmartwayZ mandate, which requires insights that are actionable at the municipal and provincial levels. A national average obscures the differences between high-performing and low-performing areas. Identifying which municipalities rank highest and lowest on both access and usage provides policymakers with a benchmark for their own area.

In Figure 3.5, the provinces are compared for the share of cycling trips within the 10-minute range of cycling. This reveals that Gelderland is the province with the highest share of cycling trips within the 10-minute bike-shed, while Groningen has the lowest share.

Next, Figure 3.6, shows that the municipalities with the highest access to the essential amenities are Haarlem, Leiden, Delft, Rijswijk, Amsterdam, 's-Gravenhage, Beverwijk, Utrecht, Oegstgeest, and Dordrecht. Which is as expected, because bigger cities have more people living close to each other, maybe even in apartment flats, and therefore it must be more dense with amenities too. So, it would be logical that these municipalities have more amenities close by. Furthermore, it shows that the ones with the lowest access rates are Nieuwkoop, Voorst, Tubbergen, Sluis, Texel, Hilvarenbeek, Midden-Drenthe, Borsele, De

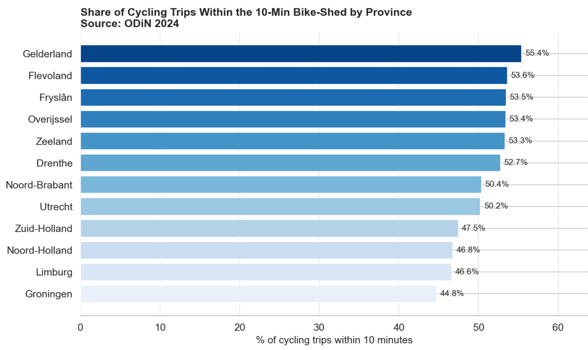


Figure 3.5: Local biking percentage per province

Wolden, and Aa en Hunze. This could be because of the contrary, these are more rural areas and therefore less dense, spread over the area. And therefore, it has fewer amenities close by.

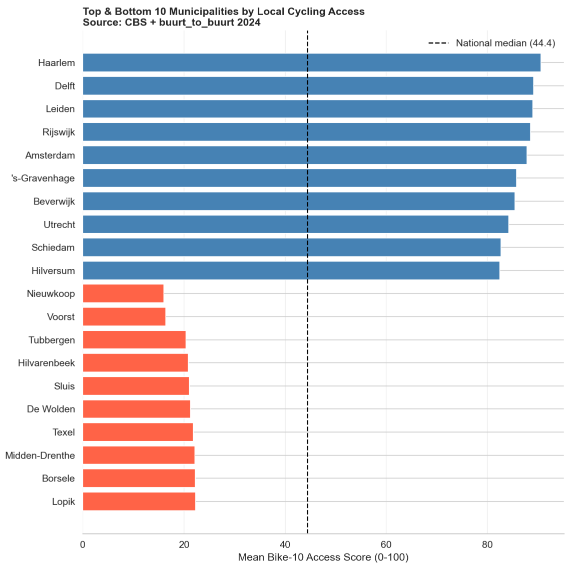


Figure 3.6: Municipality ranking based on 10-minute access

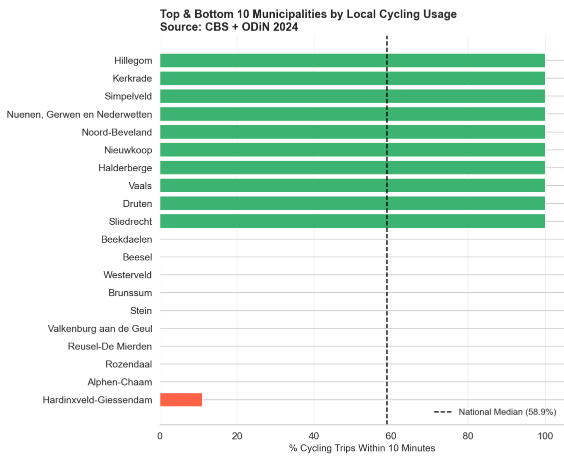


Figure 3.7: Top and bottom municipality usage

Then comes an interesting part which shows the actual cycling usage within the 10-minute access by municipality. This reveals that none of these top 10 access municipalities actually have the highest usage of the bike as transportation mode within the 10-minute access range. Furthermore, none of the lowest 10-minute access municipalities show up in the lowest cycling usage. This serves as a bridge to the next relationship that is identified. Table 3.8 lists the ten municipalities with the largest positive mismatch gap (high access, low usage) and the ten with the largest negative gap (high usage, low access), identifying the most policy-relevant cases on both ends.

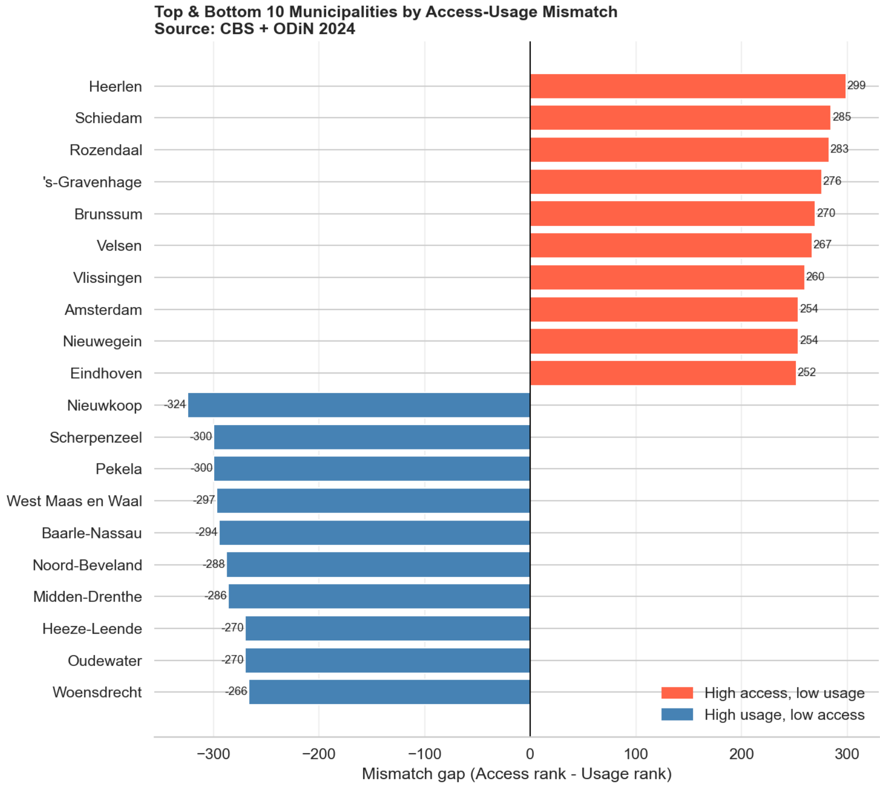


Figure 3.8: Municipality ranking top and bottom based on mismatch gaps

The access-usage gap. The second relationship concerns the mismatch between access rank and usage rank at the municipal level (mismatch_gap). This is the most policy-relevant relationship in RQ1, because it identifies two distinct types of underperformance: municipalities where spatial access is good but local cycling remains low (requiring behavioural or infrastructure interventions), and municipalities where cycling behaviour is relatively strong despite weaker access (potentially explaining where investments in amenities would have the greatest return).

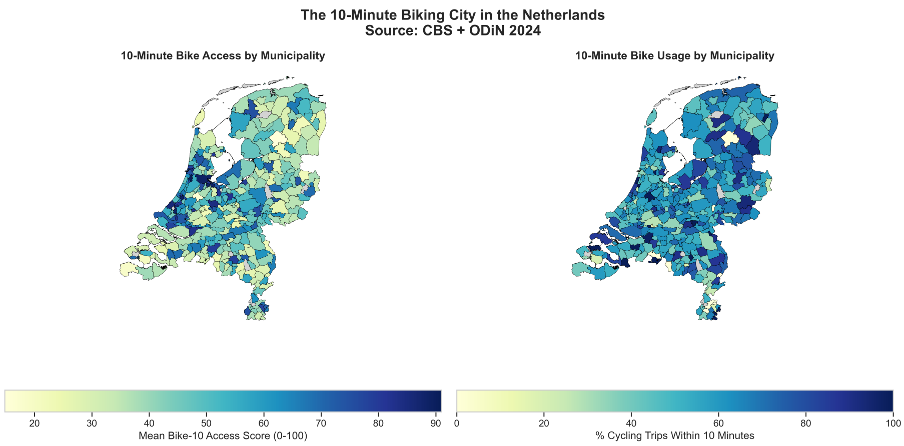


Figure 3.9: Spatial comparison of 10-minute bike access and local cycling usage by municipality

Figure 3.9 compares the spatial distribution of 10-minute bike access with local cycling usage across municipalities. This map-based comparison is useful because it shows that high access and high usage do not necessarily occur in the same places. Some municipalities may have relatively strong access to amenities but lower local cycling usage, while others may have weaker access but a higher share of local cycling trips. These differences are important for policy because they highlight distinct types of intervention needs.

Results specifically related to research question 2
Research question 2 focuses on the different income groups and how their travel modes are affected by where they come from, whether this is a strongly urban area, or not urban at all. The analysis focuses on three descriptive relationships that can be directly inspected from the available figures: income and local cycling usage, urbanisation, access, and mode choice, and finally, income and access. Income and local cycling usage focuses on whether local cycling for essential trips differs between household income groups. urbanisation, access, and mode choice describe whether accessibility and mode choice differ between very urban, moderately urban, and less urban areas. Lastly, income and access highlight whether different income groups have equal access to essential amenities within the 10-minute cycling radius.

A direct measurement of interaction between different income groups would require a separate social-exposure indicator, such as the income diversity of reachable neighbourhoods within the 10-minute cycling-shed or origin–destination income differences for actual trips. Therefore, in this descriptive section, social exposure is discussed through income-related access and usage patterns rather than as a direct interaction metric.

The following figures show how 10-minute cycling access relates to travel behaviour, income group, and urbanisation. The figures are organized according to the four relationships introduced above: distance and amenity access, income and local cycling usage, urbanisation and mode choice, and income and access. Figure B.2 in Appendix B compares the share of essential cycling trips within 10-minute across household income groups. The result appears relatively stable across the income categories, with only small differences between low-, middle-, and high-income groups. This suggests that local cycling for essential trips is not strongly concentrated in one income group; once people are already making cycling trips for essential purposes, the likelihood that those trips remain within the 10-minute cycling-shed appears broadly similar across income categories.

Furthermore, Figure B.1 in Section B shows the overall results of the share of amenities within the 10-minute isochrone. It shows which types of essential amenities are most commonly reachable within the 10-minute cycling radius. The results show that everyday services such as bus stops, primary schools, GP/doctor services, supermarkets, and pharmacies are reachable for a large share of neighbourhoods, while hospitals and train stations are less consistently available within the same radius.

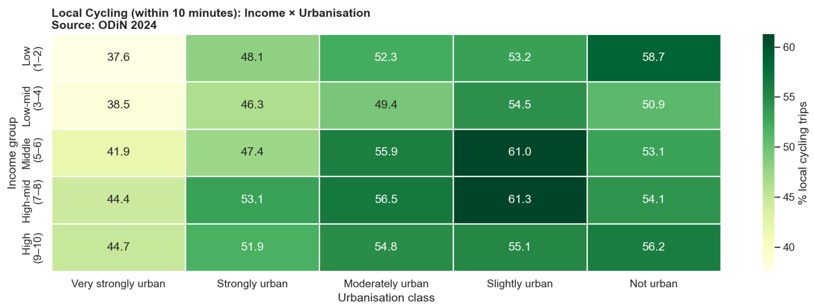


Figure 3.10: Local cycling share by income group and urbanisation class

Figure 3.10 extends the income analysis by combining income group with urbanisation class. It provides the info that two income groups can have similar national averages, but their cycling behaviour may differ depending on whether they live in very urban, moderately urban, or less urban areas. The heatmap shows whether income-related cycling patterns are actually shaped by the spatial context in which different income groups live. Diving in to the urbanisation, access and mode choice relationship, we compare the access scores across urbanisation classes.

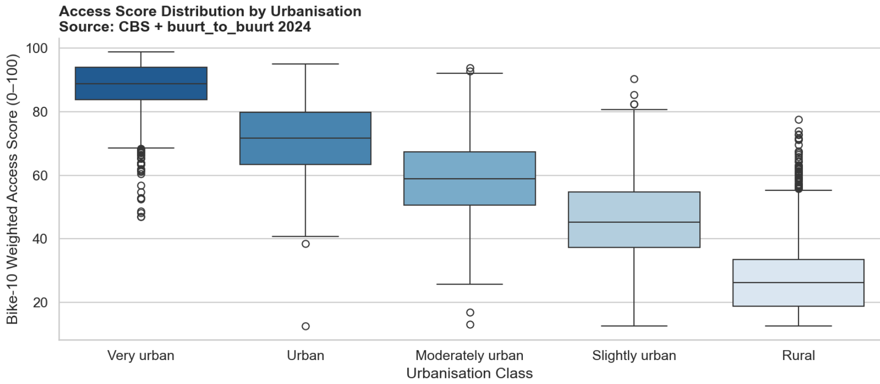


Figure 3.11: Bike-10 access score distribution by urbanisation class

Figure 3.11 shows the distribution of Bike-10 access scores across urbanisation classes. The boxplots show that accessibility depends strongly on spatial structure: more urban areas generally have higher and more consistent access scores, while less urban and rural areas show lower scores and greater variation.

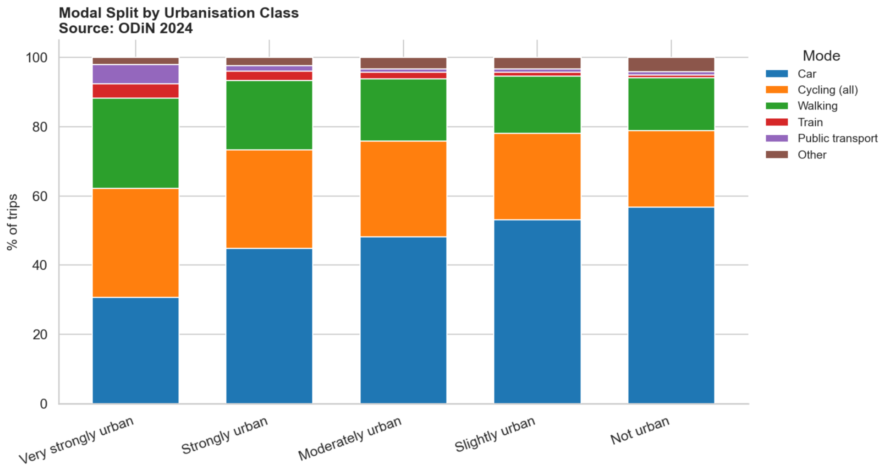


Figure 3.12: Modal split by urbanisation class

Figure 3.12 shows how the overall modal split changes across urbanisation classes. The figure indicates that car use generally increases as areas become less urban, while cycling, walking, and public transport play a larger role in more urban environments. This helps explain why amenity accessibility does not automatically translate into the same travel behaviour everywhere. In dense urban areas, nearby amenities may be supported by several non-car modes, while in less urban areas people may rely more strongly on cars even when some destinations are reachable by bike.

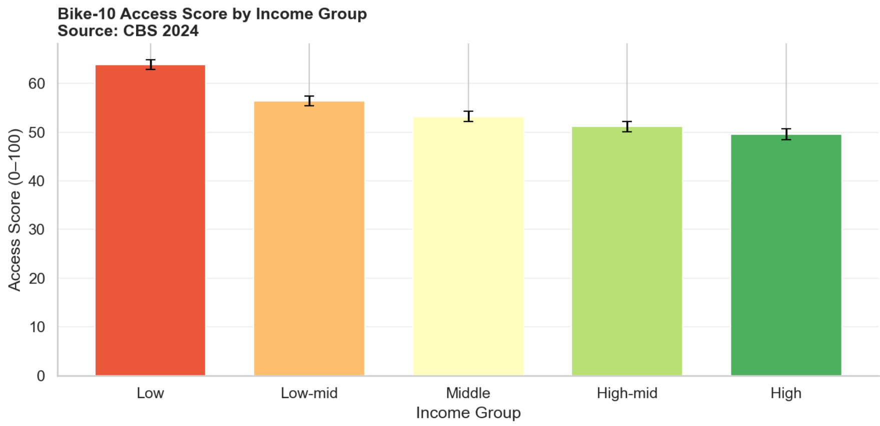


Figure 3.13: Bike-10 access score by income group

Finally, related to income and access to essential amenities, also findings are shared. Figure 3.13 compares Bike-10 access scores across income groups. This figure is central to the equity part of RQ2 because it shows whether different income groups have similar access to essential amenities within the 10-minute cycling radius. If low-income groups have lower access scores, this would suggest that the 10-minute cycling city is less equally distributed and that lower-income residents may face greater barriers to nearby essential services. If low-income groups have equal or higher access, this may reflect the fact that lower-income households are often located in denser urban neighbourhoods where amenities are closer together.

3.3 | Discussion

Overall, the results show that the 10-minute cycling city cannot be evaluated solely by the proximity of amenities. Figure 3.1 shows that cycling is strongly associated with shorter trips, supporting the use of a 10-minute cycling-shed as a meaningful local mobility threshold. Walking trips are concentrated at very short distances, cycling remains common within the local range, and car trips are generally longer. While some bicycle and e-bike trips extend beyond 3 km, the 10-minute radius remains a useful indicator of local accessibility rather than a strict behavioural limit.

The performance of the 10-minute cycling city varies between Dutch municipalities and provinces, according to the findings for research question 1. The results show that while real cycling usage and accessibility to amenities are related, they do not necessarily happen simultaneously. This means that having a lot of amenities close by does not ensure that locals will regularly opt to ride their bicycles for short trips.

Local cycling behaviour clearly varies by province, as Figure 3.5 shows. Groningen performs far worse than Gelderland, which has the largest percentage of bicycle trips within the 10-minute criterion. Spatial organization, travel culture, infrastructure quality, and alternate modes of transportation could all contribute to the explanation of these variations. Bicycle usage could be higher in provinces that have lower height differences, shorter travel distances, and strong riding traditions. But the exact reasoning of this is not found here.

Additionally, Figure 3.6 shows that highly urbanized municipalities like Amsterdam, Utrecht, Haarlem, Leiden, and Delft have the highest Bike-10 access scores. Dense urban structures that place amenities close to one another make these communities easy to visit within a 10-minute bike radius. On the other hand, because amenities are more widely distributed in rural areas, towns like Midden-Drenthe, Texel, and Aa en Hunze receive lower scores. This demonstrates that accessibility is influenced by urban density.

High accessibility does not, however, always equate to the highest bicycle usage, as Figure 3.7 shows. The municipalities with the highest local cycling utilization do not include any of the municipalities with the highest access rankings. Similarly, cities with poor accessibility may not necessarily have the lowest rates of bicycle usage. It implies that factors other than geographical access alone influence cycle behaviour. This could, for example, be the attractiveness of bicycle routes, cultural preferences, car ownership, infrastructure quality, and cycling safety, which may all have an impact on locals' decision to ride.

Figure 3.9 makes the discrepancy between cycling usage and accessibility even more evident. While some localities exhibit great cycling behaviour despite poorer accessible circumstances, others have relatively high accessibility but lower cycling utilization. Locations where policy initiatives should concentrate on enhancing cycling attractiveness, safety, or behavioural incentives are indicated by municipalities with high access but low usage. On the other hand, investments in local facilities and more direct bike connections may be more advantageous for communities with limited access but relatively high riding usage.

Overall, the results to the first research question show that the 10-minute cycling city is influenced by residents' ability and willingness to use the cycling network in addition to the geographical distribution of facilities. Therefore, depending on whether the primary obstacle is bad accessibility, inadequate cycling infrastructure, or lower-than-expected cycling usage despite strong access, municipalities need to implement different kinds of policy interventions.

For research question 2, which focuses on the different income groups and how that relates to the research, figure B.1 shows that accessibility within the 10-minute bike-shed varies by amenity type. Everyday services such as supermarkets, pharmacies, GP services, and primary schools are accessible for most neighbourhoods, whereas hospitals and train stations are less frequently reachable. This suggests that the 10-minute cycling city is better suited to daily needs than to higher-order regional services.

The income-related figures show that local cycling behaviour is not strongly concentrated in one income group. Figure B.2 shows relatively small differences in the share of essential cycling trips within 3 km between low-, middle-, and high-income groups. This suggests that, once people are already making cycling trips for essential purposes, the likelihood that these trips remain within the 3 km bike-shed is broadly similar across income categories. However, this does not mean that income is irrelevant. Rather, income should be interpreted together with other factors such as urbanisation, car ownership, destination availability, and cycling infrastructure.

Figure 3.10 shows why this combined interpretation is necessary. While income groups may appear similar at the national level, their cycling behaviour can differ once urbanisation class is included. The heatmap suggests that spatial context can shape local cycling patterns as much as, or more than, income alone. This means that policy interventions should not only target income groups in isolation, but should also consider where those groups live and whether their neighbourhoods support short cycling trips. In other words, the same income group may experience the 10-minute cycling city differently depending on whether it lives in a very urban, moderately urban, or less urban environment.

urbanisation appears to be one of the strongest contextual factors in the analysis. Figure 3.11 shows that Bike-10 access scores differ clearly across urbanisation classes. More urban areas generally have higher and more consistent access scores, while less urban and rural areas show lower scores and more variation. This is expected because dense urban neighbourhoods usually contain more amenities within a short distance, while rural neighbourhoods often have fewer nearby facilities and larger distances between destinations. Therefore, urbanisation is not only a background variable; it directly shapes the feasibility of the 10-minute cycling city.

Figure 3.12 further shows that the broader mobility environment changes across urbanisation classes. Car use generally becomes more dominant as areas become less urban, while cycling, walking, and public transport play a larger role in more urban environments. This helps explain why amenity accessibility does not automatically translate into the same travel behaviour everywhere. In dense urban areas, nearby amenities may be supported by several non-car modes, while in less urban areas, people may rely more strongly on cars even when some destinations are reachable by bike. Therefore, local access is only one condition for cycling behaviour; the surrounding transport system also matters.

The income-access result in Figure 3.13 adds an equity perspective to the analysis. It compares Bike-10 access scores across income groups and helps show whether different socioeconomic groups have similar access to essential amenities within the 10-minute cycling radius. If lower-income groups have lower access scores, this would suggest that the benefits of the 10-minute cycling city are not evenly distributed. If lower-income groups have equal or higher access, this may reflect the fact that lower-income households are often located in denser urban neighbourhoods where amenities are closer together. Therefore, this figure helps distinguish whether the 10-minute cycling city reduces inequality through strong local access, or whether it risks reproducing inequality through uneven amenity distribution.

Finally, Figure 3.9 translates the access and usage results into a spatial policy perspective. The maps show that high access and high usage do not necessarily occur in the same municipalities. Some municipalities may have relatively strong access to amenities but lower local cycling usage, while others may have weaker access but a higher share of local cycling trips. This is important because different spatial patterns require different policy responses. In low-access areas, interventions may need to focus on improving cycling connections to essential services or supporting more local amenity provision. In high-access but lower-usage areas, the issue may be less about amenity availability and more about the attractiveness, safety, or convenience of cycling compared with other modes

Some limitations to this research are that the single-year snapshot: ODiN 2024 and CBS 2024 cannot capture trends. Low respondents per neighbourhood: ODiN is representative nationally, sample sizes at neighbourhood level can be small, which might unintentionally influence the results that come out of this study, especially when looking at the lowest cycling usage plot, this revealed that there might be data missing of any usage in these places, and that could ensure that the research is focused on improving places that actually do not need most improvement. For the analysis of the income, it is good to note that the CBS income variable quality is low, as there is a high rate of missing values in the income columns. Therefore, it was already decided to look at the year 2024, instead of 2025, but including more values would generate the best results.

Future research could strengthen these findings through a longitudinal analysis using ODiN and CBS data from 2019–2024, allowing trends in accessibility and cycling behaviour to be examined over time.

3.4 | Conclusion

Taken together, the figures suggest that the 10-minute cycling city is useful as a descriptive accessibility framework, but it should not be treated as a complete explanation of travel behaviour. Nearby amenities matter, but their effect depends strongly on urbanisation, income context, the type of amenity, and the wider transport environment. The most useful interpretation of the 10-minute cycling city is therefore not simply whether a neighbourhood has a high access score, but whether that access is converted into actual local cycling behaviour across different income groups and spatial contexts. For policymakers, this means that improving sustainable mobility requires both spatial planning and mobility interventions: bringing amenities closer where access is weak, and improving the conditions for cycling where access exists but usage remains lower than expected.

4 | Predictive modelling: Mode Choice Prediction

The descriptive analysis in Topic 1 showed that cycling access and cycling usage do not always align. Some municipalities show high 10-minute cycling access to amenities, but do not necessarily show high cycling usage. Additionally, other municipalities show stronger cycling behaviour despite weaker access conditions. This suggests that accessibility alone is not enough to explain travel behaviour. Travel habits, distance, spatial and socio-economic context might also influence whether people choose to drive, use public transport, ride a bicycle, or walk. This topic extends the descriptive analysis with a predictive modelling approach. Instead of describing where access and usage differ, this analysis investigates which factors help predict travel mode choice at the trip level. This supports the policy recommendations, because these should not only show where accessibility is high or low, but also indicate which conditions are associated with sustainable mode choice.
Thus, in this section, the research question is as follows:
How do trip characteristics, travel habits, and amenity availability relate to the travel mode choice?

4.1 | Data Pre-processing

The main data source for Topic 2 is the ODiN 2024 dataset, which contains individual trip records and the transport mode. The target variable is `KHvm`, which gives the main mode class of the trip. Only regular trips were used and trips with `KHvm` values outside the six interpretable mode categories were excluded. This resulted in 172,465 usable trip records. The original `KHvm` categories were recoded into four policy-relevant classes, which can be seen in Table 4.1.

Table 4.1: Recoding of ODiN main transport mode

Original ODiN category	Recoded class
Car driver, car passenger	Car
Train, bus/tram/metro	Public transport
Bicycle	Bike
Walking	Walking

Car driver and car passenger were combined because the policy distinction of interest is mainly between car-based travel and more sustainable alternatives. Similarly, train and bus/tram/metro (BTM) were combined into public transport because both represent collective transport modes, and the separate public transport classes are smaller than car, bike, and walking, as can be seen in Figure 4.1.

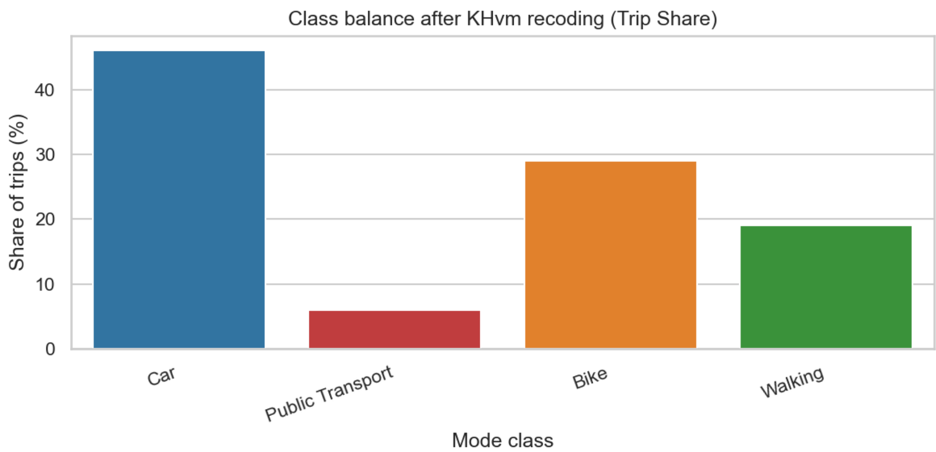


Figure 4.1: Travel mode share after recoding target variable

Initial feature selection was based on domain knowledge, and by excluding variables that would directly reveal the target outcome. The final feature set before feature selection contained 50 variables. These

can be categorised into five distinct groups: trip characteristics, individual characteristics, household characteristics, travel habits & perceived accessibility, and amenity availability.
The ODiN dataset was combined with the neighbourhood-level amenity indicators from Topic 1 using the trip origin postcode area. Since ODiN does not provide the exact origin neighbourhood, the neighbourhood amenity indicators were aggregated to the PC4 level. For each PC4, the value of an amenity variable is the share of neighbourhoods in that postcode area where the amenity is present. So if four neighbourhoods belong to the same PC4 and two of them contain a pharmacy, the PC4-level pharmacy value is 0.5. This is more informative than treating the entire PC4 as having an amenity when only one neighbourhood contains it. The trip records that could not be matched to the PC4 amenity table and other missing values were imputed using median values computed on the training set only. To prevent data leakage between the training and test set, the split was grouped by respondent ID (`OPID`). This is necessary because a single ODiN respondent can report multiple trips. If trips from the same person appeared in both the training and test sets, the model could learn a person's travel habits during training and then be evaluated on the same person's remaining trips. A `GroupShuffleSplit` was therefore used for the outer train/test split, with 15% used for testing. The resulting training set contained 146,544 trips from 45,972 respondents, while the test set contained 25,921 trip records from 8,113 respondents.

4.2 | Model Choice

To predict mode choice, two tree-based ensemble classifiers were used: Random Forest (RF) and XGBoost. These algorithms were chosen because the dataset contains a mixture of continuous variables, ordinal codes, categorical codes, and binary or proportional spatial variables. Tree-based methods can model non-linear relationships and interactions between variables without requiring strong assumptions about the underlying data distribution. Additionally, these methods do not require many pre-processing steps such as feature scaling or feature encoding (Kern et al., 2019).
Random Forest is a bagging-based algorithm that trains many deep decision trees on bootstrapped samples and averages their predictions, reducing variance compared to a single decision tree. XGBoost is a boosting-based algorithm that trains many small decision trees sequentially, where each tree focuses on correcting the errors of the previous trees. These models are often used for classification tasks and achieve strong predictive performance on structured data (Breiman, 2001; Chen & Guestrin, 2016). In transport-mode prediction, tree-based ensemble methods have also been shown to perform well on survey-based data (Tamim Kashifi et al., 2022; Wang et al., 2024).
Since both used methods are tree-based, no feature scaling was applied, keeping the original scales from the ODiN data. However, this means that coded categorical variables such as province, COROP area, or trip purpose should not be interpreted as continuous. They are used as coded controls that allow the tree models to split the data, not as variables where higher values have a meaningful linear interpretation.

4.3 | Feature Selection

Given the relatively large feature set of the dataset, a permutation importance ranking was used to identify the most predictive variables. A Random Forest (200 trees, balanced subsample class weights) was first fitted on a sub-split of the training data, and permutation importance was then computed on a held-out validation partition by measuring the drop in macro F1 when each feature's values were randomly shuffled. This approach is preferable to the default impurity-based feature importance because impurity-based methods can inflate the importance of numerical features (Pedregosa et al., 2011). By evaluating the importance on held-out data, only features that genuinely improve out-of-sample predictions are ranked highly. Figure 4.2 shows the result of this process.

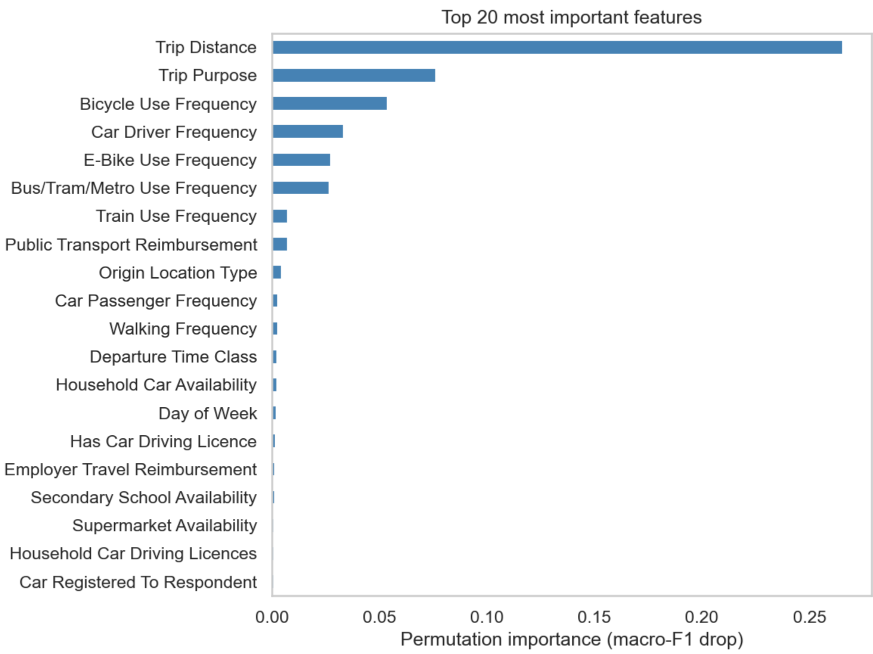


Figure 4.2: Permutation feature importance ranked by mean drop in macro F1

The feature importance results show that trip distance is by far the most important predictor of mode choice. This is expected because walking, cycling, public transport, and car trips are differentiated by trip length. Trip purpose is the second most important variable, showing that mode choice differs between commuting, shopping, education, and other purposes. Several habit-specific variables are also highly ranked. This indicates that mode choice is not only a result of the current trip, but also of the traveller's broader travel habits.

The amenity variables have a smaller permutation importance than the aforementioned variables. This suggests that the objective local amenity context contributes less to predicting mode choice than direct trip and behavioural variables. Some amenity variables, such as secondary education and supermarket availability, still appear in the top 20 most important features.

A cross-validation approach was performed to determine how many top-ranked features to keep in the final feature set. The results can be seen in Table 4.2.

Table 4.2: Cross-validation performance for top-N feature subsets

Feature subset	Accuracy	Macro F1
Top 10	0.7681	0.7446
Top 20	0.7800	0.7581
Top 30	**0.7818**	**0.7596**
Top 40	0.7802	0.7581
Top 50	0.7797	0.7573

As shown in Table 4.2, the top-30 feature subset performed best, although the improvement over the top-20 subset is small. The selected feature set is therefore the top 30 variables from the permutation-based importance ranking. These are supplemented with a small number of scenario-relevant variables, which are useful for the policy dashboard in Topic 3.

4.4 | Model Training

Baseline Random Forest and XGBoost models were first evaluated using 5-fold `StratifiedGroupKFold` cross-validation. Stratification is used to preserve the class distribution across folds, and the grouping ensures that all trips from the same respondent remain in the same fold. The baseline results are shown in Table 4.3.

Table 4.3: Baseline cross-validation performance

Model	Accuracy	Macro F1
Random Forest baseline	0.7815	0.7597
XGBoost baseline	0.7991	0.7767

As shown in Table 4.3, XGBoost outperforms Random Forest before tuning. Hyperparameter tuning was then performed with random search using 3-fold `StratifiedGroupKFold` optimizing macro F1-score. The random search focused on the main parameters controlling model complexity, regularisation, and sampling behaviour, such as tree depth, number of trees, learning rate, class weighting, and row or column subsampling. The full search space is reported in Appendix C.
Macro F1 was used as the optimisation metric because the classes are imbalanced and macro F1 gives equal importance to all mode classes. The best-tuned Random Forest achieved a macro F1 of 0.7623, whereas the best-tuned XGBoost achieved a macro F1 of 0.7769. The tuned models were then retrained on the full training data and evaluated on the held-out test set. The results are shown in Table 4.4.

Table 4.4: Final test-set performance

Model	Accuracy	Macro F1	Train-test accuracy gap
Random Forest	0.7904	0.7691	0.1654
XGBoost	0.8050	0.7838	0.0414

The Random Forest model achieved reasonable test performance, but showed clear overfitting, where its training accuracy was 0.96, while its test accuracy was 0.79. This gap indicates that the model fitted the training data too closely, and generalises less well to unseen trips. XGBoost performed better in this regard and generalised more reliably. Thus, XGBoost was selected as the model used to predict transport mode choice.

4.5 | Model Evaluation

The final XGBoost model was evaluated in more detail using class-level precision, recall, and F1 scores, a confusion matrix, and precision-recall curves. These evaluation techniques are more informative than accuracy or F1 score alone because the mode classes are imbalanced. Car and bike trips occur more often than public transport trips, so a high accuracy score alone could hide poor performance on the smaller classes. Precision indicates how often a predicted mode class is correct, while recall indicates how many trips of a true mode class are correctly found by the model. The F1 score combines precision and recall into one measure, where a higher value means a better balance between both. Macro F1 is the average F1 score across all mode classes, giving each class equal weight regardless of how many trips it contains.

Figure 4.3: Precision, recall, and F1 score by mode class for the tuned XGBoost model

Figure 4.3 shows that the model performs the best for car trips, with an F1 score of 0.85. Walking and cycling are also predicted reasonably well, with F1 scores of 0.78 and 0.76, respectively. Public transport has the lowest recall of 0.70, but still reaches an F1 score of 0.74. This is expected because public transport combines many different transport modes: train, bus, tram, and metro, which may have different predictors.

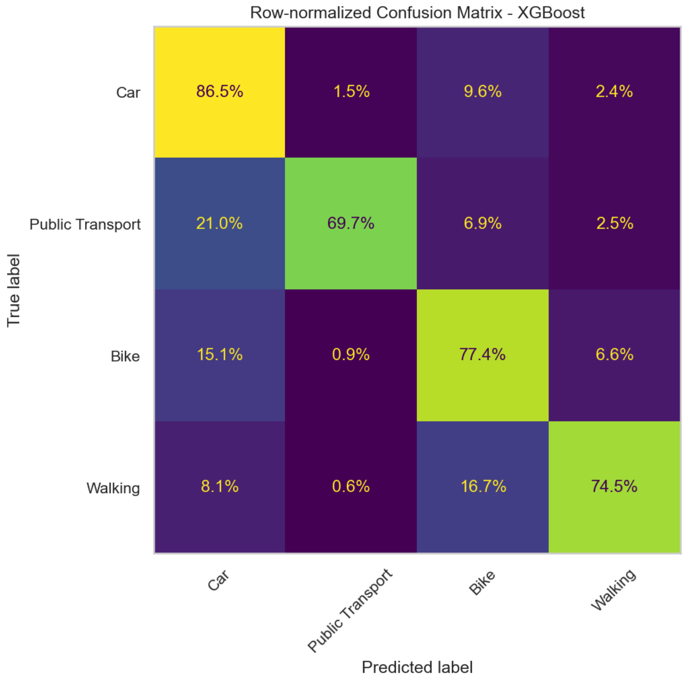


Figure 4.4: Row-normalised confusion matrix for the tuned XGBoost model

Figure C.1 provides a more detailed insight into the errors. Of all actual car trips, 86.5% were correctly predicted as car. The largest car confusion was with the bike, at 9.6%. Of all actual bike trips, 77.4% were correctly predicted as bike, while 15.2% were predicted as car. Moreover, of all actual walking trips, 74.5% were correctly predicted as walking, while 16.7% were predicted as biking. This walking-bike confusion is expected because both modes are often used for shorter trips. Lastly, public transport was correctly predicted 69.7% of the time, while 21.0% were predicted as a car. This suggests that some public transport trips share the same characteristics as car trips, such as longer distance trips.

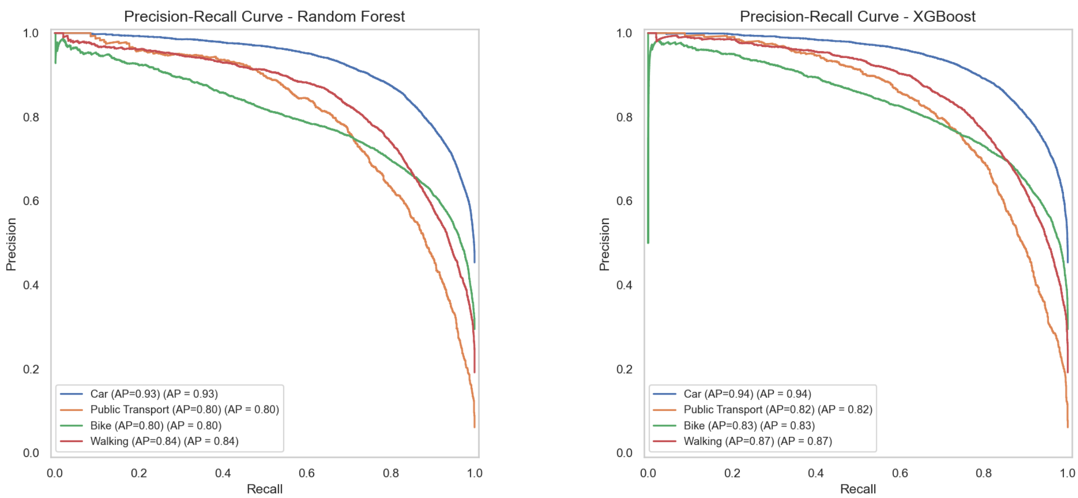


Figure 4.5: Precision-recall curves for the tuned Random Forest and XGBoost models

Lastly, Figure 4.5 supports the same conclusion. The model shows a stronger overall distinction of the classes than Random Forest, especially for the larger classes. Public transport remains the most difficult class for both models.

4.6 | Model Interpretability

To interpret which predictors most strongly influenced the final XGBoost model, SHAP values were computed for a sample of 3,000 test observations. SHAP is an explainability method that estimates how much each feature contributes to a model prediction. By averaging these contributions across many trips, SHAP shows which variables are most influential in the model overall. This makes otherwise opaque models more interpretable (Tamim Kashifi et al., 2022).

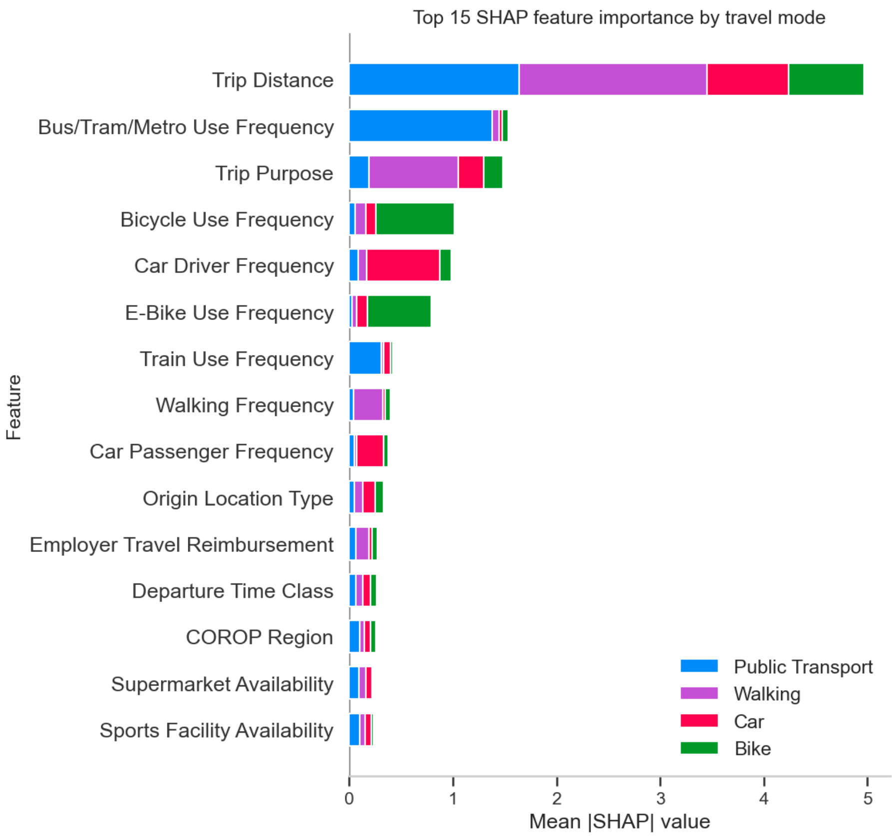


Figure 4.6: SHAP feature importance for the tuned XGBoost model

Figure 4.6 confirms the pattern observed in permutation importance. Trip distance, trip purpose, and reported travel frequencies are the strongest drivers of the predictions. This means that the model mainly distinguishes travel modes by using variables that are related to the practical feasibility and habits of using each mode. For instance, longer trips are less likely to be walking trips, and frequent bicycle users are more likely to choose the bike. The amenity variables have a smaller but still relevant contribution. It shows that amenity access is only one part of mode choice. Even when amenities are nearby, people may choose different modes depending on other variables. This aligns with Topic 1, where high amenity access did not always translate into high local cycling usage.

4.7 | Discussion

Overall, the results from Topic 2 confirm that mode choice cannot be explained by local accessibility alone. The model shows that mode choice is also related to trip distance, trip purpose, and travel habits. Accessibility creates the opportunity for cycling, but it does not automatically mean that people will cycle. Trip distance was the most important predictor. This is expected because walking and cycling are mostly used for short trips, while cars and public transport are more relevant when distances increase. However, this finding is still important because it highlights the relevance of the 10-minute cycling city. At the same time, distance alone does not determine mode choice, as people can still choose the car for short trips or the bike for longer trips. The trip purpose was another important predictor. This highlights that people do not choose their transport mode in the same way for every type of trip. Education, commuting, and shopping can all have different mode choice patterns, for instance. This is relevant for policymakers because improving access for one type of amenity might have a different effect than improving access to another. This means that for the 10-minute cycling city, it is important to look at which amenities are nearby, not just how many amenities are nearby.

Additionally, travel habits are strongly related to mode choice. People who often use the bicycle are more likely to cycle for a specific trip, while people who drive more often are more likely to use the car. This suggests the traveller's habits partly explain that mode choice. Therefore, improving access alone may not be enough to increase cycling usage. Even when amenities are local, residents may still use the car because this is what they are used to.

The amenity variables from Topic 1 were less important than the variables mentioned above. However, this does not mean that amenity availability is irrelevant. Rather, it highlights that local access is only one part of the mode-choice decision.

These findings are relevant for policymakers because different areas may require different types of interventions. In places where access is low, improving access to important amenities or creating direct cycling routes should be the focus. In places where access is already high, but cycling usage is low, the focus should be on making cycling more attractive and convenient compared to less active transport modes, such as the car.

Thus, Topic 2 adds to Topic 1 by showing why good access does not always lead to more bike usage. It should be noted, however, that the model does not prove direct causal effects, but it does show which factors are related to mode choice.

4.8 | Conclusion

The predictive model shows that transport mode choice can be predicted well using a combination of trip characteristics, traveller and household attributes, travel habits, and amenity availability. The XGBoost model performed best, with an accuracy of 80.5% and a macro F1-score of 0.78 on the test set. The main finding is that mode choice is mostly related to trip distance, trip purpose, and travel habits. Amenity accessibility also contributes, but is less important. This indicates that the 10-minute cycling city should be evaluated by more than merely looking at local amenities. Thus, Topic 2 builds on the findings of Topic 1 by explaining why good access does not always lead to high cycling usage.

5 | The urban equity dashboard: AI-powered interface

5.1 | Introduction

The goal of the project is to help policymakers with visualizations of the data to support decision-making. Section 3 established where cyling is strong or weak and how this relates to actual usage, section 4 demonstrated why individuals choose certain transport modes and which factors more strongly shape these decisions. This section translates the analytical results found in the sections into a tool that can be used to see how what-if scenarios (for instance, adding an extra education amenity) increase access and usage. This is made in the form of a dashboard. It includes an LLM-based AI agent that explains to users the effect of the what-if scenario, but can also provide other recommendations on what could be useful to improve the infrastructure and bike usage further. This is regarded as an appropriate measure, as the paper by Kalyuzhnaya et al. (2025) states that an LLM agent in city management allows for efficient processing of urban planning tasks while maintaining high relevance in responses. When using the LLM agents, the policymakers have more time to focus on decision-making, and therefore, this can lead to better accountability for improvements in challenging situations.
As interesting results were found in topics 1 and 2 regarding the low-income group and their bike usage, this topic focuses on low-income residents and helps users explore whether selected municipalities have good 10-minute cycling access to essential amenities, where low-access neighbourhoods are located, how low-income mode choice behaves in the ODiN data, and what policy actions may be useful under simple what-if scenarios. The LLM assistant does not replace the predictive or descriptive models. It acts as an explanation layer that summarizes dashboard results and gives policy-oriented recommendations based on the selected municipality and scenario.

5.2 | Method

The dashboard translates the results from Topic 1 and Topic 2 into a simple interactive tool for policymakers. The dashboard is built with Python, Streamlit is used for the interface, Pyplot for interactive graphs, Pandas and Numpy for data handling, and a Gemini API for the optional LLM-based policy assistant.
When the user runs the topic3.py in Streamlit, and the Gemini API is given, the code can find the local `'datasets/'`, and then it builds an interface with:

1. Dataset status cards showing whether the required files were found.
2. A Netherlands municipality map for selecting a municipality.
3. A fallback municipality dropdown if the map is unavailable.
4. A selected municipality summary with: Bike-10 access score, low-income share, and low-access neighbourhood share.
5. A graph explorer with original dataset views.
6. A separate what-if scenario section with updated scenario graphs.A policy assistant that explains the selected municipality and scenario.

It is important to have the following folder setup:

```
Q4-HW/
|-- topic3py
|-- datasets/
|   |-- combined_neighbourhood_dataset.csv
|   |-- Bike_Trip_purpose.xlsx
|-- data_cache/
    |-- topic3/
```

If a required file is missing, the dashboard shows a clear warning in the interface.
The interface contains:

- Two dataset bubbles, showing whether the required datasets were found.
- Dashboard controls, in the sidebar for selecting a municipality and a what-if scenario.

- Municipality summary cards, showing access score, low-income share, and low-access neighbourhood share.
- A graph carousel, where the user can move through different result graphs with Previous / Next buttons.
- A Gemini scenario assistant, at the bottom for natural-language policy questions.

The CBS KWB dataset (part of the combined dataset made in section 3) is the main spatial-access dataset. It provides neighbourhood and municipality-level information about population, income, urbanisation, and distances to essential amenities. The dashboard uses it to create a continuous bike-10-minute access score. The required data items from this dataset are:

Column	Meaning
`regio`	Neighbourhood name
`gm_naam`	Municipality name
`a_inw`	Number of residents
`bev_dich`	Population density
`p_ink_li`	Share of low-income residents
`ste_mvs`	Urbanisation code
`g_afs_hp`	Distance to GP / doctor
`g_afs_gs`	Distance to large supermarket
`g_afs_kv`	Distance to childcare
`g_afs_sc`	Distance to school
`g_3km_sc`	Number of schools within 3 km

Table 5.1: Description of dataset variables

For the bike-10-minute access score, the dashboard creates a continuous access score instead of only using a binary yes/no score. For this topic 3, as it will be a product for an end user where scores should be more understandable, having a continuous score and seeing it get updated based on the what-if scenarios was thought of to be more helpful to the user as an info. The binary score used earlier (in section 3) is thus replaced. The score is calculated from five amenity-access components:

- score_gp
- score_supermarket
- score_childcare
- score_school_distance
- score_schools_within_3km

For distance-based amenities, the logic is that when there is a 0 km distance, it results in a 100 access score. When this is 3 km or more, the access score gets a 0 value assigned. For instance, when it would be 1 km, the access score is (3-1)/3 * 100, so approximately 66.7. For the number of schools within 3 km, the score is scaled up to a maximum of 5 schools, so 0 schools get a score of 0, and 5 or more schools get a score of 100. The final `bike10_access_score` is the average of these components.
This continuous scoring method was chosen because the earlier binary method made many municipalities look like they had almost perfect access. The continuous score preserves differences between stronger and weaker access areas.
The ODiN dataset (also integarted into the combined dataset) is used to focus on the lowest-income group and inspect mode choice patterns. The dashboard filters the data to the lowest available `HHGestInkG` group and summarizes how that group travels by:

- national mode choice,
- first-10-minute trips,
- trip purpose,
- urbanisation class

Column	Meaning
`HHGestInkG`	Household income group
`Hvm`	Main transport mode
`MotiefV`	Trip purpose
`Sted`	Urbanisation class
`Prov`	Province
`Reisduur`	Travel duration
`FactorV`	Trip weight
`AfstV`	Trip distance, optional

Table 5.2: Description of travel dataset variables

The bike trip purpose file provides a lighter ODiN-derived cycling context. It is not municipality-specific, but it helps interpret cycling behaviour by trip purpose, bike type, and whether essential cycling trips are within 3 km. If this file is missing, the dashboard still works, but the cycling-purpose context graph is skipped.
The dashboard uses a carousel instead of a long list of graphs. Therefore, the user can use the created 'Previous graph' and 'Next graph' buttons to cycle through the available result views. The carousel includes the following graphs:

No.	Type	Description
1	Access-Usage Heatmap	Shows municipalities by Bike-10 access score and first-10-minute usage. It highlights: environmental success: high access and high usage, policy opportunity: high access but lower usage, and access gaps: lower access with cycling demand.
2	3 km bike-shed for selected neighbourhood	Lets the user choose a neighbourhood inside the selected municipality and displays an approximate 3 km cycling radius.
3	National access-income context	Shows where the selected municipality sits compared with other municipalities using access score and low-income share.
4	Essential function audit	Compares the selected municipality with the national average for each amenity category.
5	Low-access neighbourhood ranking	Lists the weakest-access neighbourhoods in the selected municipality.
6	neighbourhood access vs low-income share	Checks whether low-access neighbourhoods overlap with higher low-income shares.
7	Cycling-purpose context from Bike Trip file	Uses the optional Bike Trip Purpose file to summarize essential cycling trips and bike type patterns.

Table 5.3: Overview of carousel graphs

The dashboard focuses on low-income residents in two ways. From CBS KWB, it uses `p_ink_li` as the municipality and neighbourhood low-income context. And secondly, from ODiN, it filters to the lowest available `HHGestInkG` group. The ODiN low-income filter is displayed in the interface, for example, ODiN filter: lowest HHGestInkG group = 1; filtered rows = 12,828. This means the ODiN mode-choice graphs only describe the lowest-income group.
The dashboard classifies municipalities based on access and low-income context.
The current policy cases are:
At the bottom of the dashboard, the user can select a scenario from the sidebar. The options are no scenario, so it stays as it currently is, adding a grocery/supermarket, adding GP / healthcare, adding school/childcare access, improving cycling accessibility by 10%, and improving cycling accessibility by 20%. The selected scenario changes the scenario access score and policy case. The scenario model is simple and illustrative. It is used for demonstration and policy discussion, not for causal prediction.
Furthermore, the dashboard includes a Gemini-based AI policy assistant at the bottom. Example questions shown in the interface include: What if we add a supermarket to the lowest-access neighbourhoods? Should this municipality prioritize amenities or cycling infrastructure? How does the low-income focus change the recommendation? Give me a short policy recommendation for a presentation slide. The assistant receives

Policy case	Meaning
High access / lower low-income pressure	Access is strong and low-income pressure is relatively lower
High access / higher low-income pressure	Access is strong, but equity attention is still important
Low access / lower low-income pressure	Access weakness exists, but low-income pressure is less concentrated
Low access / higher low-income pressure	Priority equity case: weaker access and stronger low-income pressure overlap

Table 5.4: Policy case interpretation

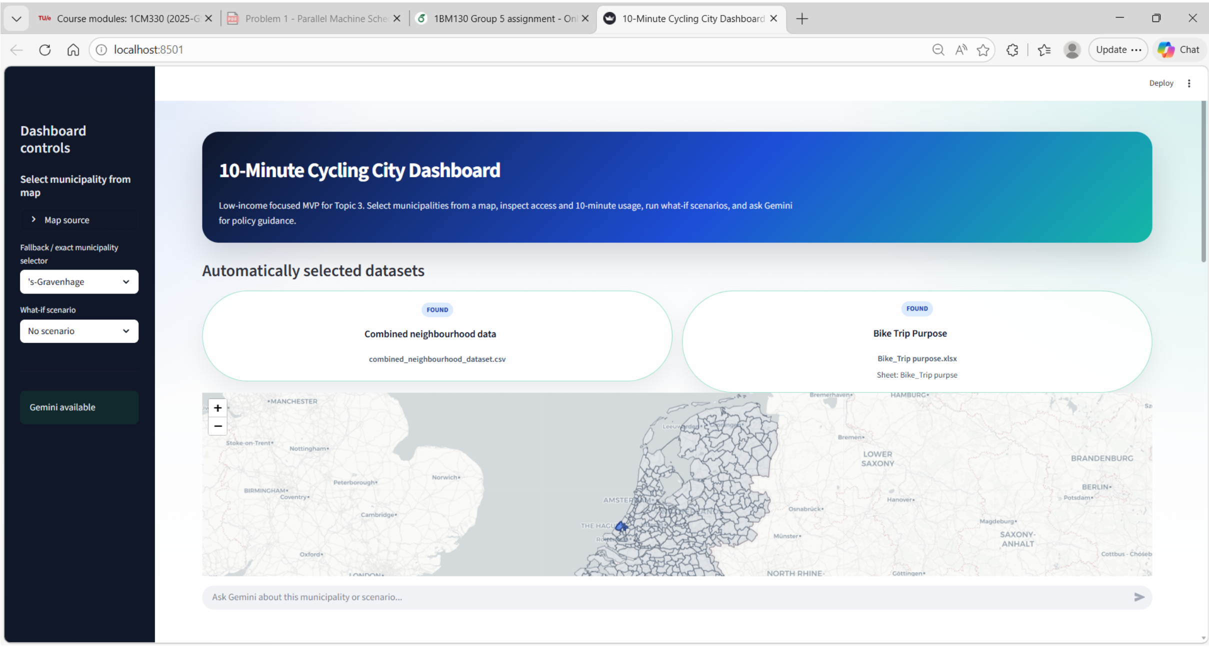


Figure 5.1: Opening page dashboard

the selected municipality, current access score, low-income share, low-access neighbourhood share, selected scenario, and scenario score. It is instructed to focus on low-income equity, avoid causal claims, avoid inventing numbers, and provide concise policy-oriented advice. If Gemini is unavailable, the dashboard uses a rule-based fallback response.
So, to conclude, for using the dashboard, one should:

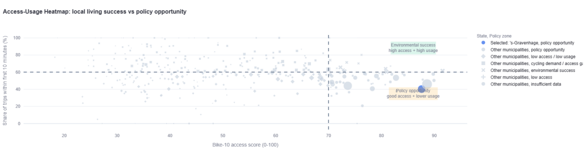
1. Put the required files in the `datasets/` folder.
2. Start the dashboard with Streamlit.
3. Check the two dataset bubbles at the top.
4. Select a municipality in the sidebar or in the graph.
5. Select a what-if scenario in the sidebar.
6. Use Previous / Next to cycle through graphs.
7. Use the Gemini assistant at the bottom to ask policy questions.

5.3 | Results

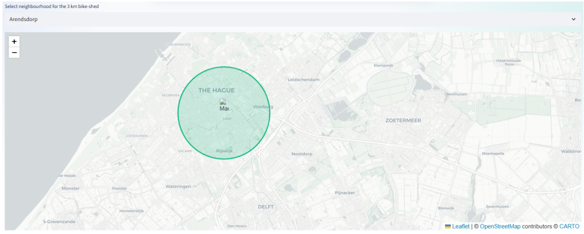
No what-if scenario

When one opens the dashboard, it shows the following, see Figure 5.1.

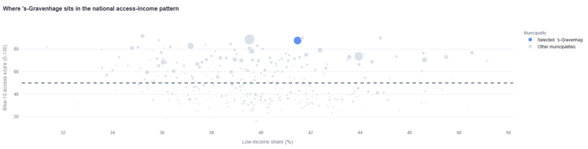
One can select a municipality by clicking on it on the map, or by selecting it in the drop-down list on the left-hand side. Then, the following graphs are shown, using the 'Previous Graph' and 'Next Graph' buttons.



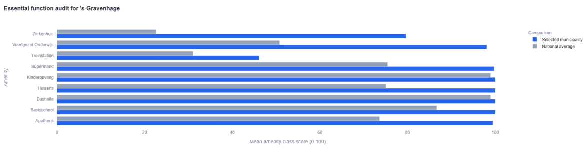
(a) Access-Usage Heatmap for 's-Gravenhage



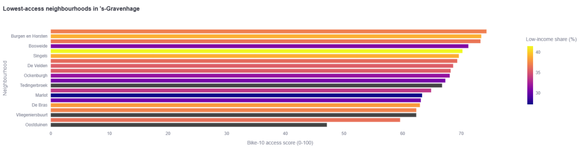
(b) 3 km bike-shed for selected neighbourhood



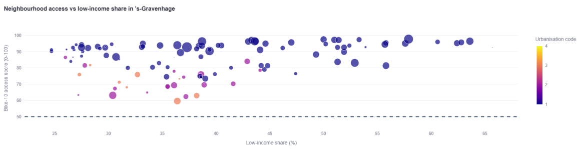
(c) Position of 's-Gravenhage in the national access-income pattern



(d) Essential function audit for 's-Gravenhage



(e) Lowest-access neighbourhoods in 's-Gravenhage



(f) neighbourhood access vs. low-income share in 's-Gravenhage

Figure 5.2: Dashboard features for assessing bicycle accessibility and equity in 's-Gravenhage. The panels show (a) access-usage heatmap, (b) 3 km bike-shed, (c) position in the national access-income pattern, (d) essential function audit, (e) lowest-access neighbourhoods, and (f) neighbourhood access versus low-income share.

For example, Figure 5.2 relates to the topic described in section 3, as it shows how the access-usage relates to each other within a municipality. It also shows how the mismatch gap exists and shows that there is an income distribution, and how that relates to the income distribution or the urbanisation class. Furthermore, it relates to section 4 by using the XGBoost model's feature importance to explain why cycling usage is high or low. The explanations are not generic; they are tailored to the selected municipality's trip composition and demographic profile. Together, this allows municipalities to test "what-if" scenarios. Where the dashboard quantifies the expected access improvements and the expected behavioral implications.
Using the drop-down list, one can select a neighbourhood, and then the map shows the accessible parts of the city within a 10-minute radius for Figure 5.2b.
In Figure 5.2a, 5.2c one can also hover over other dots with the mouse, and therefore it allows for an easy comparison across municipalities. The same applies for Figure 5.2f, only here it is on neighbourhood level. And lastly, these plots are shown, see Figure 5.3 & 5.4. These are not municipality specific.

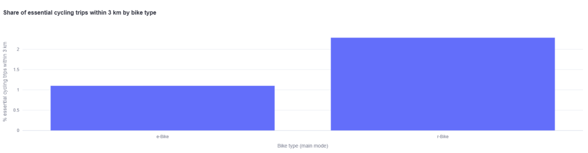


Figure 5.3: Share of essential cycling trips within 3km by bike type

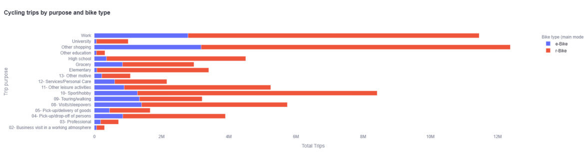


Figure 5.4: Cycling trips by purpose and bike type

Including what-if scenario

When one selects a what-if scenario on the left-hand side, the following appears when scrolled down.

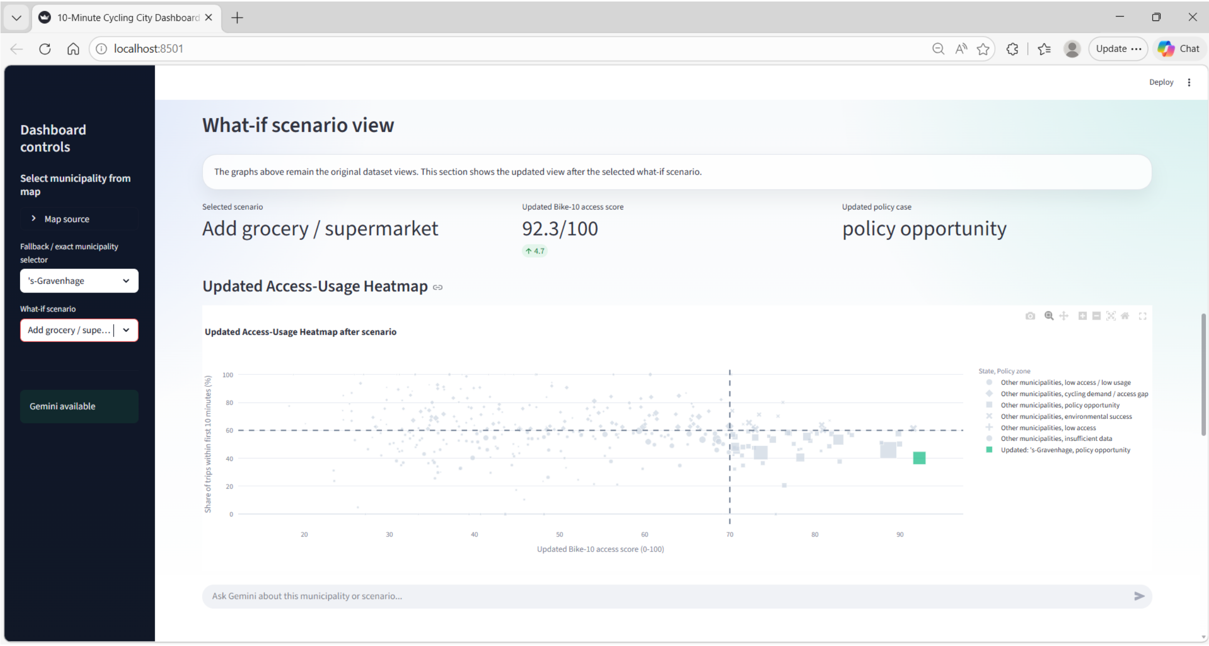


Figure 5.5: Including what-if scenario

As seen in Figure 5.5, the selected scenario is noted, and in the green arrow, the increase in accessibility score is shown. Furthermore, it prints all the plots where something has changed. Allowing one to compare the original case with what happens when this scenario is implemented.

AI agent

When scrolling further down, one can talk to the AI agent. This AI agent uses Gemini when an API key is available. If Gemini is unavailable, the dashboard falls back to a rule-based response. The agent and can only respond related to the research. Example questions are given, and here one example question is asked to the AI, see Figure 5.6.

5.4 | Discussion & Conclusion

By integrating the descriptive analytics from section 3 and the predictive modelling from section 4, the dashboard becomes more than a visualization tool, it becomes a decision-support system. The use of this dashboard allows policymakers to make better comparisons across the country and see how selected scenarios change the indicators when using the AI-powered interface. It shows the access-usage, the 3km bike-shed isochrone, the position of a municipality compared to the national levels, the essential function audit for a municipality, the lowest-access neighbourhood scores, the neighbourhood access versus low-income share, and bike trips purpose. Using this dashboard, a policymaker can thus directly compare across different municipalities. They can also use the what-if scenarios to explore how selected interventions change the dashboard indicators for a municipality. This provides insights into what a certain development does concerning the accessibility and usability of a neighbourhood or municipality. Similarly, the AI assistant can help explain the selected results and provide policy-oriented suggestions. The dashboard is useful for a policymaker as a decision-support tool because it brings the access, equity, scenario, and explanation layers together in one interface.

There are several limitations. First, the 3-km radius is shown as a circle. However, when actually using each of the separate routes in the cities, it becomes a different shape. This is the case as some of them might have a higher detour rate. Since this is not provided on the street level in the current dataset, it was impossible to visualize this in the current dashboard. It would, however, be interesting for further research to see what impact it would have on the map if a road were made straight to a certain amenity. Currently, when there is insufficient information for a certain municipality, it just drops that municipality.

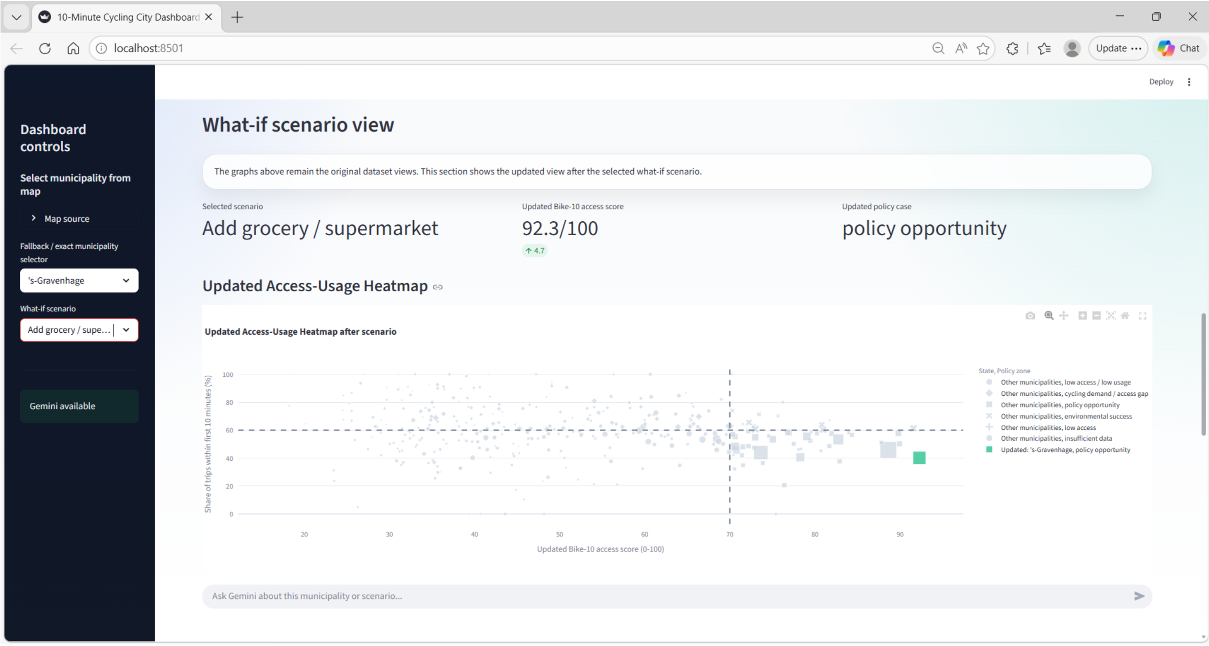


Figure 5.6: AI agent question example

It would have been useful to make a list of the municipalities that lack information, because this allows for targeting more directly at those groups, and based on that, even better advice could be given.
Future work could extend the scenario builder by calculating more intervention options, which the LLM agent can use to discuss what the best possible ideas could be used and implemented, and what the best location would be for this.
Overall, the dashboard gives policymakers a practical way to compare municipalities, identify access and equity issues, and discuss possible interventions to improve cycling access and usage.

6 | Discussion & conclusion report

6.1 | Introduction

This report investigated whether, and under what conditions, the 10-minute cycling city is a meaningful and equitable framework for sustainable urban mobility in the Netherlands. Three integrated analyses were carried out: a descriptive analysis of cycling access and usage across Dutch municipalities (Topic 1), a predictive model for transport mode choice (Topic 2), and an AI-powered interactive dashboard for policymakers (Topic 3). Together, these analyses address the SmartwayZ mandate of generating actionable, data-driven insights for municipalities developing a Sustainable Urban Mobility Plan (SUMP).

6.2 | Summary of findings

The descriptive analysis (Topic 1) showed that many cycling trips in the Netherlands fall within the 3 km bike-shed, supporting the use of the 10-minute cycling threshold as a useful local mobility concept. Cycling access, measured as a weighted Bike-10 score across fourteen amenity categories, varies strongly with urbanisation. Highly urbanized municipalities such as Haarlem, Leiden, Delft, and Amsterdam score highest, while rural municipalities score lowest. More importantly, access and usage are do not always align: none of the ten highest-access municipalities appear in the top-ten cycling usage rankings. An access-usage mismatch gap identified cities such as Heerlen, Schiedam, and Eindhoven as high-access, low-usage municipalities where behavioral or infrastructure interventions may be relevant, and municipalities such as "Nieuwkoop" and "Scherpenzeel" as high-usage, low-access areas where amenity investment would have the greatest return. Income was found to be less decisive than urbanisation in shaping cycling access: low-income neighbourhoods often have comparable or higher access scores than high-income areas, because they are more frequently located in dense urban environments where amenities are clustered.

The predictive model (Topic 2) supported the findings from Topic 1 at the individual trip level. An XGBoost classifier trained on ODiN 2024 achieved an accuracy of 80.5% and a macro F1-score of 0.78, outperforming the Random Forest model and showing less overfitting. The model predicted car, bike, and walking trips reasonably well, while public transport was more difficult to predict. Trip distance and trip purpose were the two strongest predictors of mode choice, with reported travel habit variables also ranking highly. Amenity availability contributed to the model, but ranked below trip-related and behavioural variables. This supports the interpretation that access creates the opportunity for cycling, but does not determine whether residents use it. The SHAP analysis supported the same conclusion: the model mainly distinguished travel modes through practical trip characteristics and travel habits.

The dashboard (Topic 3) translates these findings in an interactive tool for municipal policymakers, built in Python with Streamlit, Plotly, and a Gemini API integration. It combines the CBS KWB neighbourhood dataset and the ODiN travel data to visualize Bike-10 access, low-income context, and local cycling behavior for any Dutch municipality. What-if scenarios (adding a supermarket, GP service, school, or improving cycling route directness by 10-20%) update the access score and policy classification in real time. The integrated Gemini assistant provides concise, equity-focused natural-language recommendations based on the selected municipality and scenario.

6.3 | Integrated discussion

The three topics reinforce one another. Topic 1 identified the access-usage mismatch as the central policy puzzle: why do high-access areas not always cycle more? Topic 2 provided a partial answer: travel habits, trip purpose, and distance matter more than local access in determining mode choice. This means that even when municipalities improve cycling access, uptake will not follow automatically unless cycling is also made more attractive and convenient relative to car use, through measures that make cycling more attractive, safer, and easier to choose for short trips. Topic 3 translated this insight into actionable form by giving policymakers a tool that distinguishes between access-constrained municipalities (where amenity investment is most relevant) and usage-constrained municipalities (where infrastructure and behavioral interventions matter more).

The equity findings deserve specific attention. Low-income neighbourhoods do not appear to have worse cycling access on average, likely because many are located in denser urban areas. However, the results also show that income should be interpreted together with urbanisation. Similar income groups can show different cycling patterns depending on the type of area in which they live. This suggests that policy should not target income groups in isolation, but should also consider neighbourhood context, car dependence, and the local conditions for cycling.

6.4 | Limitations

Several limitations apply across all three analyses. First, all datasets (ODiN 2024, CBS 2024) are single-year snapshots. Longitudinal comparison using ODiN 2019-2024 combined with CBS 2019-2024 would allow trend detection and stronger evaluation designs. Second, ODiN sample sizes at the neighbourhood level are small, making usage estimates in sparsely sampled areas unreliable. The lowest-cycling-usage rankings in Topic 1 may partly reflect data absence rather than genuine behavior. Third, the predictive model (Topic 2) captures association, not causation. Variables such as "bicycle use frequency" are highly predictive but close to the outcome itself: habits and access can develop together over time, which a cross-sectional model cannot disentangle. Fourth, the dashboard's 3 km bike-shed is modeled as a circular radius. In practice, the reachable area follows network geometry and is constrained by detour factors that vary by neighbourhood. A full network isochrone would improve spatial accuracy.
A specific limitation of the dashboard relates to the scope of the Gemini AI assistant. The assistant was deliberately constrained to respond only within the context of the selected municipality, the dashboard's access and usage data, and the chosen what-if scenario. It was instructed not to answer general policy questions unrelated to the tool's data, not to make causal claims, and not to invent figures. This design choice was made to ensure that responses remain evidence-grounded and policymakers are not misled by AI-generated speculation. As a result, the assistant does not respond to open-ended questions outside the dashboard context (for example, general cycling policy questions or questions about municipalities not currently selected). Future work could extend the assistant's context window to include a broader evidence base, while maintaining the constraint against causal over-claiming.

6.5 | Recommendations

Based on the integrated findings, the following recommendations are made to SmartwayZ and collaborating municipalities:

6.5.1 | Differentiate intervention types by access-usage profile

Municipalities with high Bike-10 access but low cycling usage (e.g. Heerlen, Eindhoven, 's-Gravenhage) should prioritise making cycling more attractive and convenient relative to the car: cycling attractiveness, infrastructure quality, perceived safety, and short-trip car dependence. Municipalities with low access but relatively high cycling usage (e.g. Nieuwkoop, Scherpenzeel) are strong candidates for amenity investment or improved cycling connections to nearby urban centers.

6.5.2 | Target equity interventions at urban low-income neighbourhoods

The access-equality finding is partially reassuring, but urban low-income residents show lower-than-expected cycling rates despite good access. Targeted interventions should include cycling affordability support (subsidized e-bike leasing), improved safety on cycling routes through these neighbourhoods, and community-level cycling programs.

6.5.3 | Extend the dashboard with full network isochrone

The current circular bike-shed is a simplification. Integrating the SmartwayZ detour factor data into a true network isochrone would improve the spatial accuracy of access scores and make what-if scenarios more realistic.

6.5.4 | Pursue longitudinal data integration

Combining ODiN and CBS data across multiple years would allow detection of trends in access-usage alignment, enabling the tool to evaluate whether past interventions have had the intended effects.

6.6 | Conclusion

This project shows that the 10-minute cycling city is an empirically grounded and policy-relevant concept for the Dutch context, but that its impact depends critically on what lies beyond the spatial access score. Access creates the precondition for cycling; it does not create the behavior. Improving sustainable urban mobility in the Netherlands requires both place-based planning, bringing amenities within cycling distance, and mobility-focused interventions, making cycling a more attractive and realistic option for daily trips.

The combination of descriptive analysis, predictive modelling, and an AI-powered decision-support tool developed in this project provides SmartwayZ and Dutch municipalities with a practical foundation for both types of intervention.

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A | Statement AI Usage

A.1 | General use of AI

...

A.2 | Topic 1

... Claude Sonnet 4.6 was used to debug coding errors, improve visualisation clarity, and to review coding implementations. ...

A.3 | Topic 2

Claude Sonnet 4.6 and ChatGPT Codex were used as support tools during coding and writing. They were used to review the modeling pipeline, help debug coding errors, data validation & labeling, to improve the clarity of visualisations, and to export the final model. In the report, they were used to proofread and improve writing structure.

A.4 | Topic 3

...

B | Topic 1 Additional Results

B.1 | Additional variables

Table B.1: Variables used for the cycling speed and isochrone

Variable	Description
`Cycling speed`	Standard cycling speed of 18 km/h used to define the isochrone radius. E-bikes are assumed to be 25% faster, resulting in a 22.5 km/h speed and a 3.75 km radius.
`Isochrone radius (bike)`	10 minutes. Replaces the original 15-minute walking radius city concept.
`Isochrone radius (walk)`	15-minute walking radius from Abbiasov et al. (2024), used as a reference baseline to quantify the additional reach provided by cycling.
`bike_2025_smart_distance`	Network routing distance in metres between neighborhoods pairs, using smart (fastest) routing from the SmartwayZ buurt-to-buurt dataset. Used to determine which neighborhoods fall within the bike-shed.

Table B.2: Travel behavior variables used for RQ2

Variable	Description
`pct_within_3km`	Percentage of essential cycling trips that remain within the 10-minute cycling radius.
`within_3km`	Indicator showing whether a trip is shorter than or equal to 10-minute.
`mode_simple`	Simplified transport mode category, including car, cycling, walking, public transport, and other.
`is_bike`, `is_car`, `is_walk`	Mode-specific indicators used to compare active mobility with car-based travel.
`is_essential`	Indicator for essential trip purposes, including education, grocery shopping, and personal services.
`dist_km`	Trip distance in kilometers.

B.2 | Additional results topic 1 research questions 2

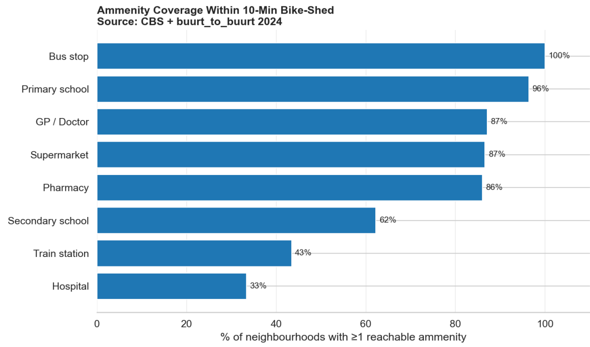

Figure B.1: Amenity coverage

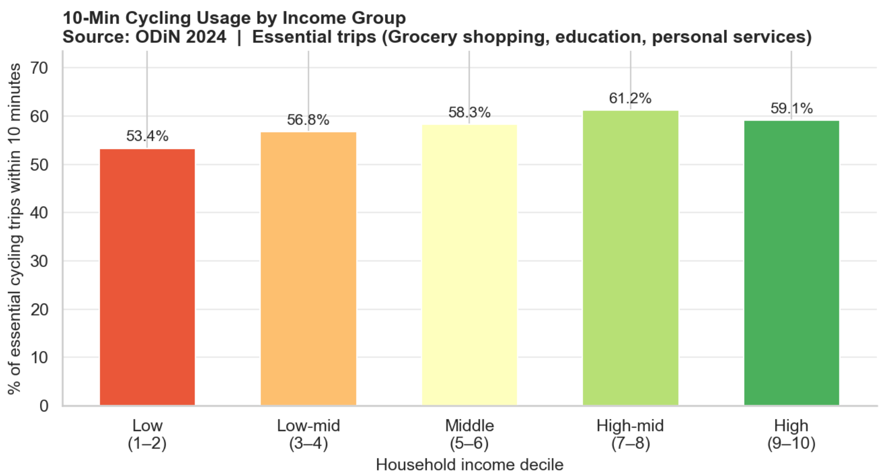

Figure B.2: Ten-minute cycling usage by income group

C | Topic 2 Additional Results

C.1 | Random Search hyperparameter tuning

Table C.1: Hyperparameter search space for Topic 2 models

Model	Hyperparameter	Values searched
Random Forest	Number of trees	200, 400, 600
Random Forest	Maximum depth	None, 10, 20, 30
Random Forest	Minimum samples per leaf	Random integer from 1 to 7
Random Forest	Maximum features	sqrt, log2
Random Forest	Class weighting	balanced, balanced subsample
XGBoost	Number of estimators	250, 400, 600
XGBoost	Maximum depth	3, 4, 5, 6
XGBoost	Learning rate	0.03, 0.05, 0.08, 0.10
XGBoost	Subsample ratio	0.75, 0.90, 1.00
XGBoost	Column subsample ratio	0.75, 0.90, 1.00
XGBoost	Minimum child weight	1, 3, 5, 8
XGBoost	L2 regularisation	1, 3, 5, 8

C.2 | Random Forest Evaluation

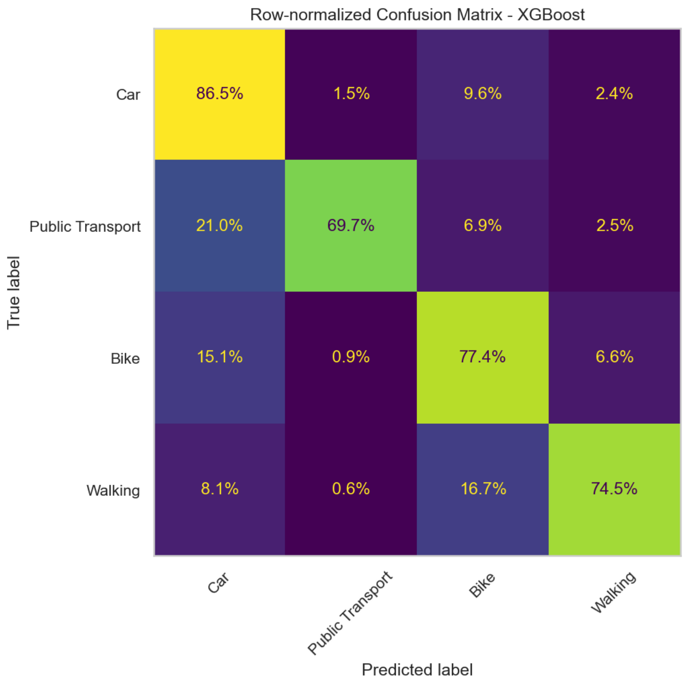

Figure C.1: Row-normalised confusion matrix for the tuned XGBoost model